Rise time of a RC circuit

1 Circuit analysis

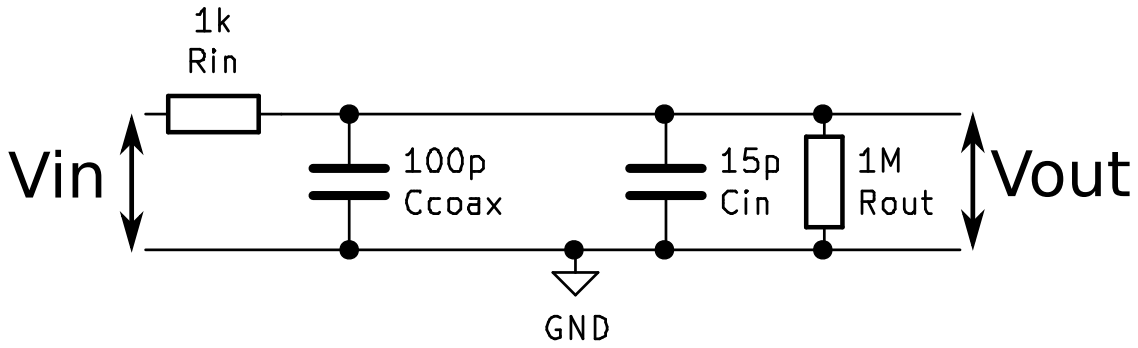


Figure 1: Equivalent model of the FG connected to the Oscilloscope using a coaxial cable.

In order to obtain the rise-time of the circuit in Figure 1 in the presence of a Heaviside function (step), first we need to analyse the circuit using the Kirchhoff laws.

As all the capacitors an in parallel, we can define an equivalent capacitance C_t as the sum of all the capacitances.

$$C_t = C_{in} + C_{coax} \tag{1}$$

To ease the analysis, we are going to define Z_e as the equivalent impedance of the parallel C_t and R_{out}, as well as applying the Laplace transformation to the capacitance: $Z_c = \frac{1}{sC_t}$.

$$Z_e = \frac{R_{osc}}{1 + sC_tR_{out}} \tag{2}$$

Once the equivalent circuit in Eq. 2 is in place, we can apply the Kirchhoff node law to the V_{out} node, and we obtain the following equation.

$$\frac{V_{in} - V_{out}}{R_{in}} = \frac{V_{out}}{Z_e} \tag{3}$$

The next step is to isolate V_{out} and as $R_{out} >> R_{in}$ we can also perform a simplification of the equations.

$$V_{out} = \frac{R_{out}}{R_{in} + R_{out} + sC_tR_{in}R_{out}} \tag{4}$$

$$V_{out}\Big|_{R_{out}>>R_{in}} = V_{in} \cdot \frac{1}{1 + sC_tR_{in}} \tag{5}$$

Once this steps are achieved, we proceed to perform the inverse Laplace transformation on the Eq 5, with the initial conditions of a Haeviside function from 0 to 1. The following equations show the $V_{out}(t)$ voltage in function of time.

$$V_{out}(t) = V_{in}(1 - \exp(\frac{-t}{\tau})) \tag{6}$$

$$\tau = C_tR_{in} = (C_{in} + C_{coax})R_{in} \tag{7}$$

2 Rise-time estimation

As we have defined in the previous steps the response of the circuit to the Heaviside function, we are in a position to obtain the rise-time of the step.

First, we define the rise-time as the time interval between the $V_{out}(t) = 10\% \cdot V_{in}$ and $V_{out}(t) = 90\% \cdot V_{in}$. In Equation 8 we can see the definition of the rise-time given this time interval.

$$t_r = t_{90\%} - t_{10\%} \tag{8}$$

Where $t_{90\%}$ is the time instant when $V_{out}(t) = 90\% \cdot V_{in}$, and $t_{10\%}$ is the time instant when $V_{out}(t) = 10\% \cdot V_{in}$.

Given the previous equations (Eq. 6, 7), we find $t_{90\%}$ in from Eq. 6.

$$\begin{aligned} 0.9V_{in} &= V_{in}(1 - \exp(-t_{90\%}/\tau)) && (9)\\ 0.1 &= \exp(-t_{90\%}/\tau)) && (10)\\ t_{90\%} &= -\tau ln(0.1) && (11) \end{aligned}$$

And we also find $t_{10\%}$ in from Eq. 6.

$$\begin{aligned} 0.1V_{in} &= V_{in}(1 - \exp(-t_{10\%}/\tau)) && (12)\\ 0.9 &= \exp(-t_{10\%}/\tau)) && (13)\\ t_{10\%} &= -\tau ln(0.9) && (14) \end{aligned}$$

Finally, we combine both equations 11 and 14, to obtain the desired rise-time.

$$\begin{aligned} t_r &= \tau ln(0.9) - \tau ln(0.1) && (15)\\ t_r &= \tau ln(9) && (16)\\ t_r &= ln(9) R_{in}(C_{in} + C_{coax}) && (17) \end{aligned}$$